CocoConnect

Project Scope Document

Client: Coconut Growers Association of Sri Lanka · Group 38

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1. Purpose

This document defines the scope of CocoConnect: what the system will do, which modules it contains, where its boundaries are, and what is left out. It is the reference for deciding whether a feature belongs in the project.

2. Project Background

Sri Lanka's coconut industry relies on phone calls, manual price negotiation, personal contacts for hiring workers, and Google Sheets for harvest records. Buyers cannot see what stock is available, prices are not transparent, and there is no way to judge whether a buyer, seller or worker can be trusted. CocoConnect brings these activities into one mobile-first platform.

3. Project Objectives

- Build a marketplace connecting coconut buyers with verified sellers and estate owners.
- Create a location-based platform for agricultural workers and service providers.
- Replace manual harvest recording with digital records and history.
- Support estate owners in managing their estates.
- Build trust between users through verification, ratings and reviews.
- Design the system so it can grow to other agricultural products.

4. Scope Statement

CocoConnect is a mobile application (Android and iOS) with a web-based administration portal, backed by a REST API and a PostgreSQL database. It serves estate owners, estate managers, buyers, sellers, workers and service providers, and administrators. The first release focuses on coconut products and coconut estate services.

5. Users

User	What they do in the system
Estate Owner	Manages estates, lists coconut harvests, responds to buyer requests, hires workers and services
Estate Manager	Runs daily work on an assigned estate and records harvests
Buyer	Browses and searches products, posts requests, negotiates prices, places orders
Seller	Creates and manages product listings and talks to buyers
Worker / Service Provider	Offers labour, machinery or other services and answers job requests
Administrator	Verifies and manages users, monitors listings, handles reports and disputes

6. Modules

The system is divided into the following modules. Each one handles one area of functionality.

Module	Main features	Scope
User Management	Registration, login, logout, password recovery, profile and profile picture, role-based access	In scope
Product Marketplace	Create, edit and remove listings; browse, search and filter; product details; contact sellers; buyer requests; price negotiation	In scope

Module	Main features	Scope
Service (Workforce) Marketplace	Service profiles; search workers and providers by type and location; job requests with accept or reject; job status	In scope
Estate Management	Register and update estates (location, acreage, trees); view estate information and statistics	In scope
Harvest Management	Record harvests; view harvest history; offline recording with later sync	In scope
Rating and Review	Rate and review after a completed transaction or service; average ratings	In scope
Notifications	Alerts for offers, messages, job requests, accepted or rejected requests and account activity	In scope
Administration	View, verify and suspend users; monitor and remove listings; review reported content	In scope
Cash / Payment Management	Recording the agreed price and total of each deal. Payment itself is made outside the system (cheque or bank transfer); online payment is a later phase	Later phase

7. System Boundaries

7.1 Inside the system

- All user accounts, roles and access control.
- Product listings, buyer requests, negotiations, orders and deal completion.
- Service profiles, job requests and their status.
- Estate and harvest records.
- Ratings, reviews and notifications inside the app.
- Administration of users, listings and reports.

7.2 Outside the system (external)

Item	How it relates to CocoConnect
Payments	Buyers and sellers pay by cheque or bank transfer outside the app. The system records the agreed price and total, but does not move money.
Delivery and transport	Arranged between buyer and seller. The system records the due date and delivered quantity only.
Maps and place search	Provided by an external map service. The system stores each user's chosen location.
SMS / push messaging	External services, planned for later phases.
Coconut quality checks	Done physically by the buyer. The system can show details such as nut weight and estate location.

8. Out of Scope

Feature	Status
Online and escrow payments	Later phase
Products other than coconut: pepper, mango, lemon, orange, pineapple, poultry	Later phase
Estate tourism: cabana booking and estate experiences	Later phase
Weather information	Later phase
Yield prediction, price trends and other advanced analytics	Later phase
AI-based recommendations	Later phase
Logistics and delivery tracking	Out of scope
Physical quality testing of coconuts	Out of scope

9. Development Approach

The project follows an Agile approach, delivered in short sprints. Each sprint produces working features that are reviewed with the client and supervisors, and their feedback shapes the next sprint. This suits the project because the requirements are still being refined with stakeholders.

Phase	Deliverable
Proposal milestone	Requirements, scope, architecture, proposal
Sprint 1	SRS, use cases, database design, wireframes
Sprint 2	UI/UX design and frontend
Sprint 3	Authentication, marketplaces, harvest module
Interim milestone	System integration and presentation
Sprint 4	Estate management and admin portal
Sprint 5	Analytics, notifications and offline support
Sprint 6	Testing and optimisation
Final milestone	Documentation, deployment and final presentation

10. Assumptions and Constraints

10.1 Assumptions

- Users have a smartphone and basic skills to use it.
- Users give correct information when registering and listing.
- Most functions need an internet connection; harvest recording should also work offline.

10.2 Constraints

- Built by a student team within two semesters.
- Low or no running cost: free tiers and open-source tools are preferred.
- User data must be protected, and each role may only use its own functions.
- The interface must support English, Sinhala and Tamil.

11. Scope Change Control

New feature requests from the client are recorded and checked against this document. A request that fits an in-scope module is added to the next sprint. A request outside the scope is added to the later-phase list and discussed with the supervisors before it is accepted.